

Neoplasia Examination

(Based on the 7th & 8th Lectures)

Part I: Multiple Choice Questions (MCQs)

1. Metaplasia is defined as:

- A) Disordered proliferation of epithelial cells.
 - **B) Transformation of one differentiated cell type into another differentiated cell type.**
 - C) Increase in cell size causing organ enlargement.
 - D) Spread of tumor cells to distant organs.
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2. Barrett's esophagus is an example of metaplasia in which:

- A) Columnar epithelium changes into squamous epithelium.
 - **B) Squamous epithelium changes into intestinal epithelium.**
 - C) Normal cells become undifferentiated malignant cells.
 - D) Fibrous tissue is replaced by fatty tissue.
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3. Dysplasia is best described as:

- A) A benign tumor of surface epithelium.
- **B) Disordered epithelial cell proliferation considered precancerous.**
- C) A detached circulating intravascular mass.
- D) Tumor invasion of the basement membrane.

4. Lipoma is a:

- A) Benign tumor of glandular epithelium.
 - B) Malignant tumor of mesenchymal origin.
 - **C) Common benign tumor of fatty tissue.**
 - D) Locally aggressive bone tumor.
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5. One of the most common sites for blood (vascular) metastasis is:

- A) Heart and spleen.
 - **B) Bone, brain, lung, and liver.**
 - C) Muscle and skin.
 - D) Salivary glands and pituitary gland.
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Part II: Essay & Comparison Questions

1. Classification of Tumors According to Tissue of Origin

Tissue Type	Tissue Origin	Benign Tumor	Malignant Tumor
Mesenchymal	Fatty tissue	Lipoma	Liposarcoma
	Smooth muscle	Leiomyoma	Leiomyosarcoma
Parenchymal (Epithelial)	Gland	Adenoma	Adenocarcinoma

Tissue Type	Tissue Origin	Benign Tumor	Malignant Tumor
	Skin surface	Squamous cell papilloma	Squamous cell carcinoma

2. First Three Microscopic (Cytologic) Criteria of Malignancy

1. **Increased nucleus-to-cytoplasm (N/C) ratio**
 2. **Hyperchromatism** (darkly stained nuclei)
 3. **Pleomorphism** (variation in size and shape of cells and nuclei)
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3. Microscopic Picture of Leiomyoma

- Composed of **interlacing bundles of smooth muscle fibers** cut in different directions.
 - Cells are **long and spindle-shaped**.
 - Nuclei are **centrally located, blunt-ended, and cigar-shaped**.
 - Cytoplasm is **abundant and eosinophilic**.
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4. Comparison Between Benign and Malignant Tumors

Feature	Benign Tumor	Malignant Tumor
Rate of growth	Slow	Rapid
Capsule	Usually present	Usually absent
Mode of growth	Expansion & compression	Infiltration & invasion

Feature	Benign Tumor	Malignant Tumor
Metastasis	Absent	Frequent
Differentiation	Well differentiated	Poorly differentiated

5. Routes of Spread of Malignant Tumors

1. **Direct spread** (local invasion).
2. **Lymphatic spread.**
3. **Vascular (hematogenous) spread.**
4. **Transcoelomic spread** (e.g., Krukenberg tumor).
5. **Perineural spread.**